

EYES IN THE THICKET: A NEURAL GAZE FOR INVISIBLE WILDLIFE ENCOUNTERS

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Abstract—ResNet50-driven wild animal detection systems represent a significant leap toward harmonizing ecological surveillance with intelligent computation. This paper introduces a real-time monitoring solution that leverages convolutional deep learning and embedded vision to identify elusive wildlife in complex environments. As human expansion interfaces more frequently with natural habitats, the urgency to detect and prevent wild animal incursions has escalated. Traditional systems falter under dynamic conditions—low light, camouflage, partial visibility—wherein animals blend seamlessly with the background. Our system addresses this through residual learning blocks capable of preserving low-level spatial cues critical for animal detection amidst occlusions and background clutter. Powered by a live feed pipeline and optimized using transfer learning, the model adapts to diverse species with minimal retraining effort. The GSM-integrated alert system transmits notifications instantaneously upon detection, offering a non-intrusive, proactive response layer. Deployment scenarios range from farm borders to remote forest outposts. Comparative experiments benchmark ResNet50 against YOLOv3, SSD, and Fast R-CNN, demonstrating superior accuracy and latency balance. This system further integrates energy-conscious hardware (Raspberry Pi and thermal cameras) to ensure long-term, low-power operation in rural terrains. Our research contributes not just a technological framework but a shift toward ethical, responsive coexistence between humans and wildlife. The proposed system acts as both observer and guardian—an unblinking neural gaze embedded within the thicket.

Keywords—ResNet50, Real-Time Detection, Transfer Learning, Camouflage Recognition, Deep Convolutional Networks, GSM Alert System, Edge AI, Wildlife Surveillance, Neural Feature Extraction, Embedded Vision System

I. INTRODUCTION

The dynamic interface between expanding human settlements and shrinking wildlife habitats has intensified the need for robust, intelligent systems that can effectively detect, monitor, and mitigate wild animal intrusions. With the rapid urbanization of forest-adjacent zones, the probability of encounters between humans and wild species

has sharply increased, leading to damaging consequences for both parties. Wild animal intrusions into agricultural fields and rural habitations result in crop destruction, livestock attacks, and in some cases, human fatalities. Conversely, retaliatory action from humans often endangers the very existence of wildlife, disrupting ecological balance and endangering biodiversity. This pressing concern has catalyzed the evolution of wildlife surveillance into a technology-driven domain, underpinned by advances in machine learning, edge AI, and intelligent vision systems [1], [2].

Traditional approaches such as manual patrolling, infrared trap cameras, and fence-based detection systems fall short in delivering real-time alerts and reliable species identification. These methods are constrained by delayed human response, limited monitoring range, lack of continuous data streams, and inability to function efficiently under low-light or obstructed environments. The evolution of intelligent surveillance—especially deep learning-enabled real-time detection systems—has emerged as a transformative solution in addressing these inefficiencies [3], [4]. Recent developments have introduced transfer learning and hybrid deep neural network models, which significantly reduce training overhead while boosting detection accuracy. For instance, a hybrid architecture involving VGG-19 and Bi-LSTM has shown the capacity to classify animal activity patterns with 98% accuracy and to issue predictive alerts through automated messaging systems [2]. Such models not only enhance the immediacy of threat identification but also support intelligent alert dissemination, reducing response latency in critical scenarios.

Another pivotal frontier in this domain lies in edge computing integration. Systems embedded with real-time detection algorithms operating on compact devices such as Raspberry Pi and NVIDIA Jetson Nano have demonstrated feasibility for remote deployments with constrained power

supply [3]. This edge-AI paradigm alleviates the need for continuous cloud connectivity, enhancing both speed and security in field operations. In parallel, deformable convolution layers and multi-scale feature extraction introduced in modified YOLOv2 frameworks have improved spatial recognition under diverse environmental conditions, including occlusions and camouflage scenarios [4].

Energy efficiency also forms a cornerstone in the design of deployable wildlife detection devices. Intelligent systems employing multi-modal sensors—such as motion detectors, infrared modules, and low-resolution thermal cameras—have been optimized with algorithmic strategies like motion sensitivity control, frame-difference preprocessing, and modular activation patterns. These configurations yield a reduction of over 50% in energy consumption, proving essential for off-grid conservation setups [1]. Equally important is the rising role of few-shot learning and meta-learning in addressing the challenge of data scarcity for rare or camouflaged animal species. Few-shot frameworks such as FS-CDIS (Few-Shot Camouflaged Detection and Instance Segmentation) leverage limited annotated samples to enable generalizable, high-precision recognition, thereby filling a critical void in wildlife image analysis [5]. These models deploy innovative mechanisms like instance triplet loss and instance memory storage to distinguish visual foregrounds from complex, naturally cluttered backgrounds—effectively countering the problem of camouflage.

YOLO-based networks continue to dominate the real-time object detection landscape due to their streamlined, single-shot architecture. Optimizations like the improved YOLOv5s variant and modified YOLOv7 have been explored for detecting large wildlife on roadsides, especially in scenarios involving fast motion, variable lighting, and occluded views [6], [10]. These advancements are critical for integration into autonomous vehicle systems, forest patrol drones, and mobile surveillance towers. The incorporation of attention mechanisms and thermal image fusion further elevates performance in nocturnal detection, as demonstrated by frameworks employing heat map-based localization [9].

Moreover, context-aware detection systems are being coupled with smart alert modules that rely on GSM, LoRa, or Wi-Fi-based communication to broadcast warnings to predefined user groups—ranging from forest officials to local farmers. When combined with geolocation data and temporal behavior analytics, these alerts serve as early intervention tools that can prevent physical confrontations and mitigate property damage [2], [3], [8]. Despite these significant strides, several challenges persist. Models trained on conventional datasets often fail when exposed to real-world wildlife images plagued by illumination inconsistencies, motion blur, and environmental occlusions. Model generalization remains a core issue, particularly in deploying detection systems across varying terrain and climatic zones. Moreover, balancing computational load with inference speed on embedded devices continues to test architectural efficiency [4], [7].

The demand for ultra-lightweight yet accurate detection pipelines remains high, urging ongoing innovation in model pruning, quantization, and knowledge distillation techniques.

In response to these multifaceted needs, this research introduces a real-time wild animal detection framework titled —Eyes in the Thicket: A Neural Gaze for Invisible Wildlife Encounters. This system employs a ResNet50-based deep convolutional neural architecture fine-tuned via transfer learning, integrated with real-time camera feeds and GSM alert functionalities. It is specifically designed for detecting camouflaged or fast-moving animals in low-light, occluded, and dynamic environmental conditions. The proposed system merges high detection accuracy with low-latency alert dissemination, all while operating within energy-efficient embedded hardware boundaries. In essence, this paper contributes to the field in three impactful dimensions: first, by proposing a robust detection framework capable of tackling camouflage and motion variance; second, by ensuring operational viability through compact edge computing; and third, by enabling autonomous alert propagation to safeguard human settlements and support conservation objectives. This intersection of intelligent vision, ethical AI, and wildlife protection marks a transformative leap in sustainable coexistence strategies for rapidly changing ecological boundaries.

II . LITERATURE SURVEY

Wild animal detection has witnessed substantial advancement in recent years through the integration of energy-efficient systems, deep learning, and intelligent alert mechanisms. In [1], an energy-conservative strategy was presented, incorporating motion sensor sensitivity calibration and functional segmentation within hardware modules, ultimately reducing the system's power demand by over half. This approach is crucial for long-term field deployment, especially in off-grid forest borders. Another model in [2] integrated a hybrid deep neural network architecture, combining convolutional feature extraction with temporal analysis to classify animal behavior in real time. The model achieved high classification accuracy and enabled automated alert dissemination, enhancing proactive threat mitigation. Work in [3] addressed both detection and deterrence by leveraging embedded edge-AI platforms coupled with ultrasound emission tailored to specific species. This not only detected wildlife but actively repelled them, offering a dual-layered security approach. A lightweight detection system was proposed in [4] using a restructured YOLOv2 model embedded with deformable convolutions, resulting in optimized accuracy and inference time on low-resource devices.

The challenge of detecting well-camouflaged or rarely seen species was approached in [5], where a few-shot learning mechanism was introduced alongside a novel triplet loss-based segmentation framework. This enabled detection in low-data environments, addressing one of the major gaps in wildlife recognition. In [6], a YOLOv5s-based model was optimized for forest environments, improving

detection efficiency for large animals across diverse lighting and occlusion conditions. A classical CNN-based approach in [7] showcased a standard classification pipeline but lacked adaptability under complex terrain features. Enhancing upon this, [8] proposed a hybrid system combining traditional convolutional models with LSTM-style memory units to improve sequential prediction in intrusion events. A further advancement in model adaptability and performance under occlusion is seen in [9], where attention-based mechanisms were applied within a deep learning framework for superior localization. The latest advancement in [10] introduced a modified YOLOv7 variant explicitly designed for animal detection in outdoor scenes, outperforming earlier models under challenging environmental constraints.

Ref	Model/Technique	Key Feature	Energy Efficient	Application Scope
[1]	Frame Diff. + Sensor Sensitivity	4-tier energy optimization	✓Yes	Long-term deployment, low-power zones
[2]	VGG-19 + Bi-LSTM	Hybrid spatial-temporal detection	Moderate	Real-time forest surveillance
[3]	YOLO + Edge AI + Ultrasonic Repeller	Detection repulsion embedded platforms	✓Yes	Smart agriculture, crop protection
[4]	Optimized YOLOv2 with DCL	Fast lightweight embedded devices	✓Yes	Edge deployment, real-time detection
[5]	FS-CDIS + Triplet Loss	Few-shot camouflaged animal segmentation	Moderate	Rare animal detection, low-data scenarios
[6]	Improved YOLOv5s	Forest-optimized large animal recognition	Moderate	Roadside wildlife detection
[7]	CNN Standard Classifier	Baseline deep learning pipeline	Low	General animal detection
[8]	CNN + LSTM (DeepAID)	Smart intrusion classification with memory	Moderate	Wildlife-intrusion prediction
[9]	TMS-YOLO with Attention	Attention-enhanced detection & localization	✓Yes	Occlusion-prone terrain
[10]	Modified YOLOv7	Optimized for complex outdoor environments	✓Yes	Road & forest environment detection

TABLE 1 : Comparative Analysis of Reviewed Models

III. PROPOSED METHODOLOGY

The proposed system, titled “*Eyes in the Thicket*”, is a modular and scalable real-time wild animal detection framework that leverages a ResNet50-based deep convolutional neural network, edge-level computation, and automated GSM alert mechanisms. This architecture is designed to detect wildlife presence from live video streams while operating under low-power and real-world constraints, making it suitable for deployment in forest-adjacent zones and agricultural fields.

3.1 SYSTEM ARCHITECTURE OVERVIEW

The system is composed of five interdependent modules: (1) Input Acquisition, (2) Image Preprocessing, (3) Deep Feature Extraction and Detection, (4) Alert Communication, and (5) Data Logging. The model executes on a compact embedded board (e.g., Raspberry Pi 4) interfaced with a high-resolution infrared-capable camera. All components are encased in a weather-resistant enclosure for environmental resilience.



Figure 1: Prototype implementation of the proposed wild animal detection system

The physical realization of the proposed wild animal detection system is structured around a modular embedded design, optimized for real-time inference, portability, and energy efficiency. The central processing unit of the system is a Raspberry Pi microcontroller board, mounted securely on a base PCB. This board acts as the command center, orchestrating the flow of data from sensory input to output signaling. Attached to the system is a 16x2 character LCD display module, which serves as an immediate user interface, presenting real-time status updates. In the displayed configuration, the message "ANIMAL DETECTION" confirms successful identification of motion-triggered wildlife presence based on model inference. The detection pipeline is triggered by external motion sensors (typically PIR-based), which activate the processing flow only when movement is detected, thereby conserving energy during idle periods. Once triggered, the system captures a frame from a connected night-vision or infrared camera (not visible in the image but assumed connected via GPIO/UART). The frame is processed through a pretrained ResNet50 model deployed within the Raspberry Pi, which performs classification to determine animal presence.

For communication and alert transmission, a SIM800L GSM module is integrated into the circuit. This module transmits SMS alerts to preconfigured mobile numbers, ensuring that forest officials or local farmers are instantly notified. The presence of a red LED indicator adjacent to the module provides a visual cue for successful transmission or detection events. The system also incorporates a UART converter for stable communication between serial interfaces and modules, preventing data loss during runtime. Power to the entire system is supplied through a 12V DC input, regulated to meet the operational requirements of each module. The wiring architecture and

component placement suggest a focus on compactness and weather resistance, making it ideal for field deployment in rugged or remote environments. All components are neatly fixed on a base plate with drilled mounts, highlighting attention to structural stability for long-term operation. This physical architecture mirrors the proposed algorithmic flow: motion sensing → image capture → preprocessing → model inference → alert generation → status display. The simplicity of the build combined with the complexity of the AI logic enables this system to be both technically robust and practically deployable in real-world conflict zones between humans and wildlife.

3.2 IMAGE ACQUISITION AND PREPROCESSING

Live camera feeds are continuously captured and filtered using frame differencing to detect motion. Detected frames are resized to 224x224 resolution, normalized, and color-corrected. In low-light conditions, histogram equalization is employed to improve visibility, and Gaussian filtering is applied to suppress environmental noise. These techniques ensure that the image data fed into the deep neural model is clean, contrast-enhanced, and computationally efficient.

3.3 DEEP LEARNING-BASED DETECTION

At the core of the detection mechanism lies a **ResNet50** deep convolutional neural network, pre-trained on ImageNet and further fine-tuned using a curated wildlife dataset. The architecture retains residual connections that preserve gradient flow across deep layers, allowing for effective learning of complex patterns such as camouflage and partial occlusion. After the image passes through the ResNet50 layers, the extracted features are evaluated using a softmax classifier that determines whether an animal is present and optionally identifies its species. Frames with detection probabilities above a threshold are flagged for downstream processing.

3.4 ALERT GENERATION AND COMMUNICATION

Once an animal is detected with high confidence, the system triggers the GSM module to send an SMS alert to a predefined set of mobile numbers. The alert message contains details including the detection timestamp, animal type, and location if GPS data is available. An onboard 16x2 LCD screen provides real-time local feedback with text such as —ANIMAL DETECTION. The GSM triggering mechanism is interrupt-driven to minimize latency between detection and alert transmission.

3.5 SYSTEM INTEGRATION AND ENERGY OPTIMIZATION

Given that this solution is designed for deployment in remote or power-constrained environments, energy efficiency is a key feature. The system uses a solar-charged battery unit and incorporates low-power scheduling strategies. During idle periods, the Raspberry Pi enters standby mode, and the detection pipeline is activated only when motion is detected via a passive infrared (PIR) sensor. Additionally, the neural model is optimized for edge inference using quantization and lightweight tensor formats, ensuring minimal energy draw during computation.

3.6 DEPLOYMENT STRATEGY

The prototype is deployed in forest border zones, near agricultural plots, or animal crossing points. It is housed in a waterproof casing and mounted securely at an elevated position to maximize field of view. Periodic maintenance is supported through modular SD card updates or wireless synchronization if connectivity is available. Field testing under variable conditions (day/night, open/occluded areas) has validated the system's reliability, with low false positives and robust operational stability.

IV. EXPERIMENTAL SETUP AND RESULTS

The implementation and evaluation of the proposed system, “*Eyes in the Thicket*,” were conducted within a controlled and semi-field environment to ensure realistic validation of its performance metrics. The experimental design focused on the critical parameters of accuracy, real-time responsiveness, power efficiency, and deployment scalability. The system was engineered to simulate wild animal intrusion scenarios in agricultural and forest-edge zones while maintaining continuous operability under varying environmental conditions.

4.1 HARDWARE CONFIGURATION

The embedded platform used for the prototype consisted of a Raspberry Pi 4 Model B (4GB RAM) operating under the Raspbian OS. It was connected to a 5MP night-vision-enabled camera module, ensuring visibility in both daylight and low-light scenarios. A SIM800L GSM module was interfaced for wireless alert transmission, while PIR motion sensors facilitated intelligent trigger-based activation to conserve energy. Power was supplied through a 12V DC battery system augmented by a solar panel, enabling uninterrupted, sustainable operation in off-grid environments.

Peripheral components included a 16x2 LCD module to display detection events locally, a UART serial converter for communication stability, and external SD storage for boot and model loading. The complete assembly was housed in a weather-resistant acrylic casing and mounted at an elevation of approximately 1.5 meters to optimize camera view angles and reduce false activations caused by ground-level noise such as vegetation movement.

4.2 SOFTWARE AND MODEL CONFIGURATION

The core detection model was built using Python, with support from TensorFlow, OpenCV, and supporting libraries. The backbone model architecture was ResNet50, initialized with pretrained weights from the ImageNet dataset and fine-tuned using a domain-specific dataset consisting of 5,000 labeled wildlife images. This dataset included common forest-dwelling species such as deer, wild boars, leopards, elephants, and cattle. To increase robustness and generalization, a variety of data augmentation techniques were employed—such as horizontal and vertical flips, brightness shifts, zoom ranges, and random rotations.

The image inputs were standardized to 224×224 resolution and normalized to a [0,1] range. The model was trained using a categorical cross-entropy loss function, optimized with the Adam optimizer at a learning rate of 0.0001. Early stopping and dropout layers were applied to avoid overfitting. Post-training, the model underwent quantization to reduce its memory footprint and allow efficient edge-level deployment without sacrificing detection precision

4.3 DETECTION FLOW AND TRIGGERING

A real-time pipeline was implemented wherein the PIR sensor served as the primary wake trigger for the processing loop. Upon motion detection, the system captured a snapshot frame, pre-processed it (resized, denoised, contrast-enhanced), and passed it to the ResNet50 network for classification. If the softmax probability exceeded 85%, the GSM module was activated to send an SMS alert to a list of registered numbers. The message typically included metadata such as timestamp, location (if GPS-enabled), and classification label.

Simultaneously, the event was displayed on the LCD module in textual format ("ANIMAL DETECTION"). The end-to-end latency, measured from PIR trigger to SMS dispatch, averaged 2.7 seconds, ensuring real-time responsiveness. The system maintained 5–6 FPS throughput under normal lighting and reduced to 3–4 FPS during low-light operation due to additional image processing steps.

4.4 EVALUATION METRICS

The system was evaluated using a test dataset of 1,000 unseen animal images. Metrics analyzed included Accuracy, Precision, Recall, F1-Score, and Inference Speed (FPS). Performance was benchmarked against YOLOv3 and Fast R-CNN models on the same dataset and hardware configuration.

Metric	ResNet50 (Ours)	YOLOv3	Fast R-CNN
Accuracy	96.45%	93.80%	90.55%
Precision	94.10%	91.20%	88.60%
Recall	95.75%	90.80%	87.90%
F1-Score	94.92%	91.00%	88.24%
FPS (Speed)	5.2	8.5	3.6
Energy Draw	Low	High	Moderate

Table 2 : Comparison of Proposed System

While YOLOv3 delivered a higher FPS, it compromised on precision and recall, particularly in environments with occlusion or camouflage. The proposed ResNet50 model demonstrated higher detection consistency and superior F1-score, making it better suited for scenarios where accuracy is paramount.

4.5 FIELD DEPLOYMENT AND OBSERVATIONS

A prototype was field-tested at a farm located adjacent to a known wildlife corridor in Tamil Nadu, India. The system

was operational for 10 continuous days, logging real-world detection events. A total of 23 animal entries were successfully recorded, which included wild boars, spotted deer, and stray cattle. Only 2 false positives were encountered—both triggered by dense foliage movement under strong wind conditions. SMS alerts were received on average within 3 seconds of detection, with all detection events properly logged. The solar-powered battery system ensured continuous availability, with peak daytime charging sustaining night time operations. The night-vision capability, enhanced through histogram equalization, allowed effective detection under minimal lighting. PIR-based motion activation proved effective in conserving system resources and preventing unnecessary computation during idle periods. The figure 2 below illustrates an example of cheetah identification using our advanced system. The system successfully recognizes the cheetah in the image with a precision of 97.64%, demonstrating the model's high accuracy in distinguishing cheetahs from other animals or backgrounds. This result highlights the robustness of the algorithm in real-world wildlife monitoring applications.



Figure 2: Cheetah - Wild Animal Detection System

Actual: Cheetah
Prediction:Cheetah

4.6 COMPARATIVE ASSESSMENTS

In a comparative deployment, YOLOv3 was observed to struggle with cluttered backgrounds and performed inconsistently at night, often requiring GPU-level hardware for real-time inference. Fast R-CNN, while more interpretable, exhibited high inference times and excessive memory usage. The proposed ResNet50 framework, with its transfer learning backbone and edge optimization, offered a balance between processing efficiency and classification strength.

Furthermore, the incorporation of GSM alerting and modular sensor activation made the system fully autonomous—capable of functioning without internet connectivity or human intervention. This makes it a viable candidate for large-scale deployment in forest reserves, wildlife buffer zones, and remote agricultural plots.

4.7 LIMITATIONS

While the system showed strong operational stability, a few limitations remain. During foggy or rainy conditions, camera visibility dropped significantly, resulting in blurred

frames and occasional misclassification. Additionally, the model was less accurate when exposed to animal species not included in the training dataset. Although data augmentation helped improve generalization, edge cases involving rapid motion or partial visibility remained challenging. Another limitation was the dependency on PIR sensors. Fast-moving smaller animals such as foxes or rodents occasionally escaped detection either due to insufficient IR signature or frame-skip artifacts. To overcome this, integration with audio sensors or multiple angle-cameras could be explored in future revisions.

In summary, the experimental setup validated the proposed system's primary objectives—high detection accuracy, low energy consumption, and reliable real-time alerts. Through integration of deep learning, embedded vision, and communication technology, “*Eyes in the Thicket*” presents itself as a practical, deployable, and scalable solution for mitigating human-wildlife conflicts. The field deployment demonstrated resilience, low false positives, and effective alert delivery. Combined with edge AI optimizations, this system contributes a forward-looking perspective to wildlife conservation, surveillance automation, and sustainable coexistence in ecologically sensitive zones.

CONCLUSION AND FUTURE ENHANCEMENTS

The proposed system, —Eyes in the Thicket, offers a significant advancement in the domain of intelligent wildlife detection and early alert mechanisms. Built upon a ResNet50 deep convolutional architecture, the system bridges the technological gap between wildlife behavior understanding and real-time response capability. Through the integration of PIR motion sensors, live camera feeds, adaptive preprocessing techniques, and GSM-based communication, the system delivers reliable detection even in challenging lighting and environmental conditions.

Experimental results demonstrated that the system achieved high accuracy (96.45%) while maintaining energy efficiency and acceptable processing latency, making it viable for off-grid deployments in ecologically sensitive areas. Comparisons with existing models such as YOLOv3 and Fast R-CNN highlighted the superior performance of the proposed system in detecting camouflaged and occluded animals under field conditions. The seamless fusion of edge computing and lightweight neural inference not only ensured autonomy but also contributed to sustainable surveillance operations.

Despite its effectiveness, the system exhibited limitations under fog, rain, and species generalization. These factors emphasize the importance of further research into multi-sensor data fusion, including acoustic and thermal modalities. Future work will focus on incorporating self-learning and federated learning modules to allow adaptive model updates from distributed deployments. Additionally, we plan to integrate GPS mapping and LoRa-based wide-area alerting to facilitate region-wide threat broadcasting. Ultimately, this work paves the way for ethical and proactive coexistence strategies that safeguard both human interests and wildlife integrity in shared environments.

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